SELF-TIMED MOS CURRENT MODE LOGIC FOR DIGITAL APPLICATIONS

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ABSTRACT

This paper describes a Self-Timed MOS Current-Mode Logic (ST-MCML) for digital applications. The architecture and operation of ST-MCML is explained and analyzed. 4-bit ripple and 16-bit Carry Look Ahead adders are implemented using the ST-MCML technique in a 0.18μm 1.8V 1-GHz CMOS process. ST-MCML is compared to conventional MCML, static CMOS and domino logic in terms of power, delay, Power-Delay-Product (PDP) and Energy-Delay-Product (EDP). ST-MCML achieves low-power values as well as minimum PDP and EDP values.

1. INTRODUCTION

The VLSI development goals are high integration density, high-speed operation, low power dissipation and low cost VLSI's. In order to achieve these requirements, simple process, small area, small signal swings and low supply voltage circuits are needed. Most of these goals can be obtained from improving process technology, such as shrinking devices. However, low-power high-speed processors will require new circuit techniques, than by just improving process technology. A logic style that is becoming increasingly popular is MOS Current Mode Logic (MCML)[1],[2]. Its high-speed switching and reduced output voltage swing contribute to its high-performance. Furthermore, MCML optimization has many more degrees of freedom in the parameter selection [3]. This has been a key advantage over conventional CMOS designs as process technology scales down. However, one of the most serious drawbacks associated with MCML is the continuous static power dissipated by the current source.

A method should therefore be devised in order eliminate this static current, while attaining the domain of freedom in the parameter selection. This paper proposes a Self-Timed MCML (ST-MCML) design that allows the current source to only switch when inputs are present. Otherwise, it is OFF. This eliminates static currents when the circuit is not operating. Furthermore, the proposed implementation does not employ a voltage bias generator, as conventional MCML designs. Consequently, no special tree is needed for the reference voltage distribution, which facilitates the complexity of the layout stage.

A brief overview of the operation of conventional MCML will be first given. The proposed Self-Timed MCML is then analyzed, identifying its components. 4-bit ripple and 16-bit carry-look-ahead adders are implemented in ST-MCML. The designs are then compared with conventional MCML, static CMOS and domino logic styles. A conclusion to this work is drawn at the end.

2. CONVENTIONAL MCML

Figure 1 shows an MCML inverter/buffer gate. The gate employs a DC current source controlled by a voltage reference V_{bias}, while R are pull up resistors. The logic function is implemented by the logic block connected between the resistors and the current source. For an inverter/buffer, the logic block is the differential pair where the input signals enter.

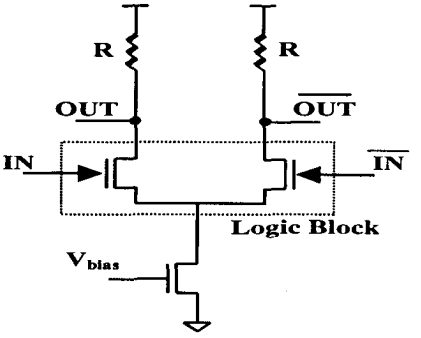


Fig. 1. Conventional MCML Schematic

The operation of the MCML logic is based on the differential pair circuit. The value of the input controls the current flow through the two branches. The amount of current passing through the ON branch controls the discharge delay of the logic gate ($1 \rightarrow 0$ transition), while the load resistor controls the charging of the output nodes ($0 \rightarrow 1$ transition).

The output voltage swing V_{swing} is defined as the voltage difference across the discharging load resistor at steady state. The small output swing of MCML circuits reduces the cross talk between adjacent signals. On the other hand, a small V_{swing} is not recommended because it causes the gate to be more susceptible to noise and reduces the current difference between the differential branches, which produces a smaller discharge current. Normally a swing 20% of V_{dd} is used [1]. MCML circuits also have noise immunity, due to its differential nature.

If properly designed, MCML transistors must not operate in the cut-off mode independent of the input combination [4]. MCML thus only operates in the saturation and linear modes for fast switching.

MCML has some major drawbacks which limit its use in digital systems. First is its static power dissipation due to the constant current source which is independent of the operating frequency. Therefore, MCML is preferred in high frequency applications only, in order to reduce the overhead of its static biasing power. Furthermore, MCML is not suitable for power-down modes because of the DC current source. MCML designs also employ a reference voltage distribution tree to control the current source of each gate, leading to a larger chip area and more complex routing.

With these drawbacks in mind, a Self-Timed MCML design is developed. The proposed design eliminates the mentioned drawbacks associated with conventional MCML while achieving lower Power-Delay-Product values. A description of the proposed ST-MCML design is first given, followed by a performance comparison with other

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logic styles.

3. SELF-TIMED MCML

Designs implemented in Self-Timed MCML (ST-MCML) employ a pulse generator and sense amplifiers to sense the output signals. Instead of being controlled by a constant V_{bias}, the gate of the MCML's current source is controlled through pulses generated from the pulse generator as shown in Figure 2. The generated pulses supply current for a period of time to the gate when logic is evaluated. This eliminates any unnecessary static power being dissipated in conventional MCML. Small voltage drops are established at the output of the evaluating gate. These voltage drops are sensed by sense amplifiers, and full rail logic is produced at the sense amplifier output.

3.1. Pulse Generator

To generate the different timing signals for the gates and sensing amplifier, an on-chip pulse generator is needed, which detects a change in the input signals. Figure 2 shows the schematic diagram of the pulse generator which consists of an AND gate and a delay element with delay = τ_d.

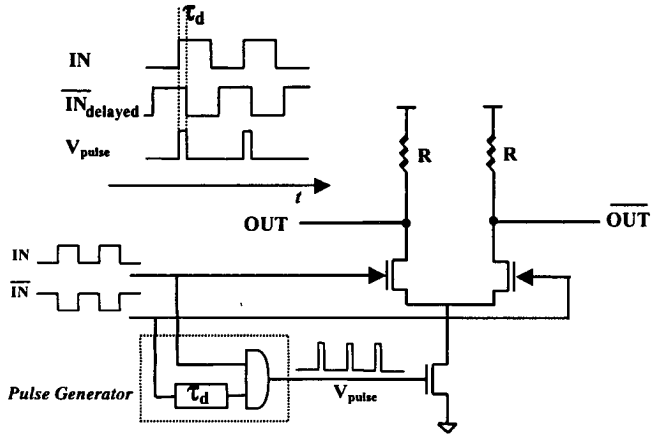


Fig. 2. ST-MCML Schematic

Short pulses are generated when a transistion in the input occurs (LOW to HIGH or vice versa). The overall pulse width is controlled by the delay element. A pulse width of 100psec (τ_d=100psec) from a 1nsec period (Input frequency=1GHz) is chosen after optimizing the design for power and delay, as well as robustness. An important point to consider is that the current source will remain OFF, and the gate will dissipate minimum power while latching the correct data if no transistion occurs, whether it is because 1. No input data is available or 2. Input data does not change. For example if the input to ST-MCML is (0000111000), ST-MCML dissipates power only when transition takes place from "0" to "1", and from "1" to "0" again. On the other hand, no static power is dissipated if a bit is followed by a similar bit in the next cycle. Meanwhile, conventional MCML designs dissipate power continuously independant on the input pattern.

3.2. Sense Amplifier

The main function for the sense amplifier is to sense the small voltage drops that are established at the output of the evaluating gate (IN and $\overline{IN}$). Consequently, full rail logic is produced at the sense amplifier output ($Sense_{Out}$ and $\overline{Sense_{Out}}$).

The differential sense amplifier/latch used on a number of Alpha microprocessors [5] has been chosen in the implementation of the ST-MCML design. Figure 3 shows the used sense amplifier.

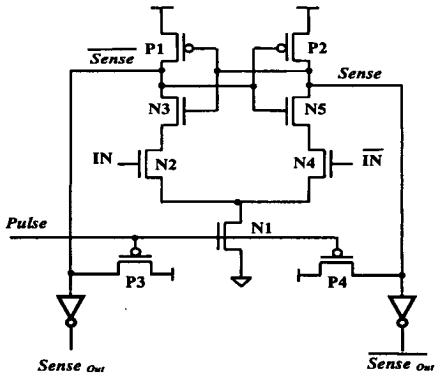


Fig. 3. Sense Amplifier Schematic

The employed sense amplifier consists of a pair of NMOS devices (N2 and N4) which are used to sense the final voltage difference from the output of the logic blocks which are the inputs to the sense amplifier (IN and $\overline{IN}$). A single NMOS pulldown device (N1) is used to enable the sense amplifier by providing current pulses produced by the pulse generator. A cross-coupled inverter pair (P1/N3 and P2/N5) in a cascode configuration is used to amplify the detected voltage differential. Finally, a set of inverters are used to buffer the output of the sense amplifier.

Operation of the sense amplifier begins with the Pulse signal in a LOW state. The precharge devices (P3 and P4) set the $Sense$ and $\overline{Sense}$ nodes to V_{dd}. The preceeding logic blocks achieve a small differential voltage on the sense amplifier input (IN - $\overline{IN}$). Meanwhile the sense amplifier is held in the precharge state while the small differential is achieved on the sense amplifier inputs. Consequently, the pulldown device N1 to turn ON, supplying current to the two legs of the sense amplifier.

In the design of the sense amplifier, many factors must be considered. During the phase of layout, matching between the two legs of the sense amplifier is required to avoid introducing offsets that would effect sensitivity, while the sizing of the device N1 is a balance between speed and sensitivity [6].

4. CARRY LOOK AHEAD ADDER

To demonstrate the functionality of ST-MCML, a 16-bit Carry Look Ahead (CLA) adder was chosen as a test vehicle because of its general structure, and that it employs various gates with different fanouts. Four 4-bit CLA adders cascaded in a ripple adder configuration, construct the 16-bit CLA adder.

Figure 4 shows the architecture of the 4-bit CLA adder implemented in ST-MCML. The pulse generator, a 100psec delay element, the 4-bit CLA adder and the sense amplifiers are shown.

The 100psec delay element is added for the following reason: Opening the sense amplifier prior to building sufficient differential voltage exposes any noncommon-mode noise, device mismatches, and could possibly lead to incorrect data recovery. The generated pulse is therefore asserted after a time period (100psec) to allow sufficient dif-

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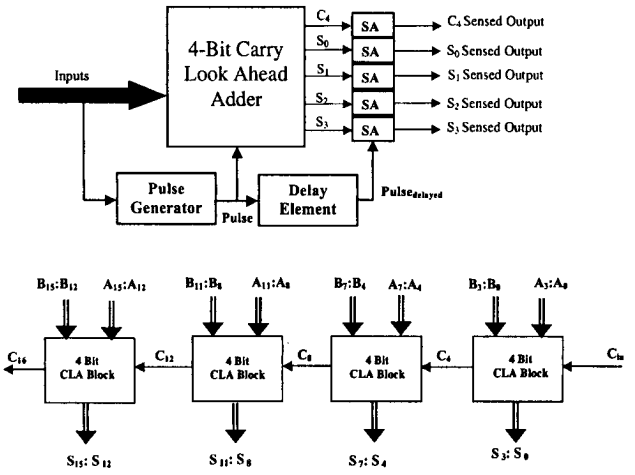


Fig. 4. ST-MCML CLA Adder

ferential to build up. Therefore, the pulse enabling the sense amplifier ($Pulse_{delayed}$) is a delayed version of the original pulse signal generated by the pulse generator (Figure 4). The $Pulse_{delayed}$ signal is delayed by 100psec to give time for the logic gates to evaluate the correct data. It should also be noted that all gates of the CLA adder, having different depths are supplied by the original single pulse generated. This is because the pulse has a width of 100psec which is sufficient to activate a circuit with a delay not exceeding 100psec (which is the case here). A circuit delay over 100psec, requires a delayed version of the pulse, which was the case for the sense amplifiers. As shown in Figure 4, all logic gates in the 4-bit CLA adder are pulsed using the pulses produced directly by the pulse generator, while the sense amplifier is enabled by the delayed version of these pulses.

Figure 5 illustrates a generate gate as an example for complex logic gates in the 4-bit CLA adder.

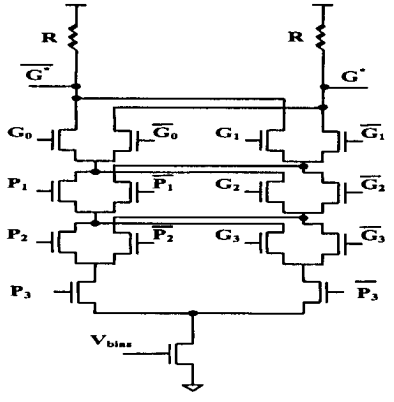


Fig. 5. Generate gate of the 4-bit CLA adder

The gate utilizes the exclusive relationship between propagate and generate signals to reduce the number of transistors and the gate complexity. The logic expression for this gate is

$$G^{\star} = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 \qquad (1)$$

The waveforms produced to generate the output carry bit; $C4$ and the most significant sum bit; $S3$ are shown in Figure 6.

Seven waveforms are shown. The third waveform shows the delayed pulse that supplies the sense amplifier. The first shows the $S3$ and $\overline{S3}$ bits with a 300mV differential. The differential $S3$-$\overline{S3}$ to be sensed is shown in the sixth waveform. Similarly, the seventh waveform shows the $C4$ and $\overline{C4}$ bits with a 220mV differential (shown in fifth waveform). Finally, waveforms 2 and 4 show the sensed full rail $S3$ and $C4$ bits at the sense amplifier output.

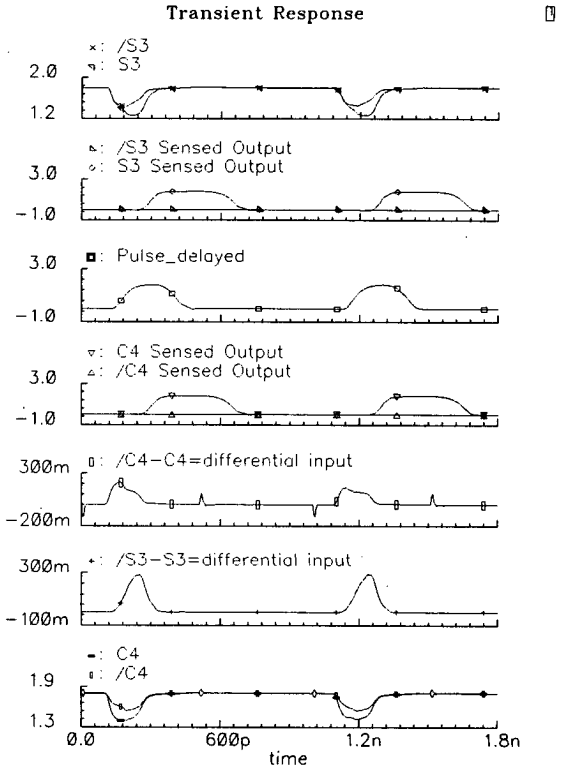


Fig. 6. Generation of the S3 and C4 bits

Ideally, the voltage levels at the input of the sense amplifier should be V_{dd} and V_{dd}-ΔV, where ΔV is the voltage drop to be sensed which is analogous to the V_{swing} value of the ST-MCML gate ≈300mV. In reality, the high voltage is not V_{dd} as seen in the first and last waveforms in Figure 6. It experiences a voltage drop, because practically, little current flows in the OFF branch of the MCML gate. However, as long as there is a sufficient differential voltage applied to the sense amplifier inputs, correct data is evaluated.

The 16-bit CLA adder as well as a 4-bit ripple adder were also implemented in conventional MCML, static CMOS and domino logic. A comparison between these designs will follow. Figure 7 shows the Sum and Carry circuits that construct the full adder (FA) in MCML, which is the same implementation as ST-MCML when the current source is pulsed by the pulse generator.

Table 1 summarizes the comparison between different implementations of the 16-bit CLA adder and the 4-bit ripple adder in terms of delay, power, Power-Delay-Product (PDP) and Energy-Delay-Product (EDP). All values are normalized to those of static CMOS, and the switching activity was assumed to be 50%. The reported delay is the

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Table 1. Comparison between logic styles

Logic Style	4-bit Ripple Adder (Normalized)				16-bit CLA Adder (Normalized)			
	Delay	Power	PDP	EDP	Delay	Power	PDP	EDP
CMOS	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Domino	0.67	4.10	2.75	1.84	0.67	2.50	1.68	1.12
MCML	0.27	3.70	1.00	0.27	0.42	4.20	1.76	0.74
ST-MCML	0.47	1.10	0.52	0.24	0.65	1.20	0.78	0.51

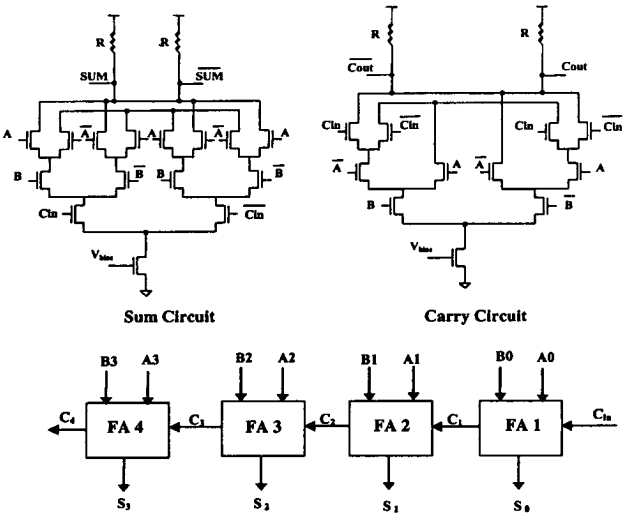


Fig. 7. MCML Ripple Adder

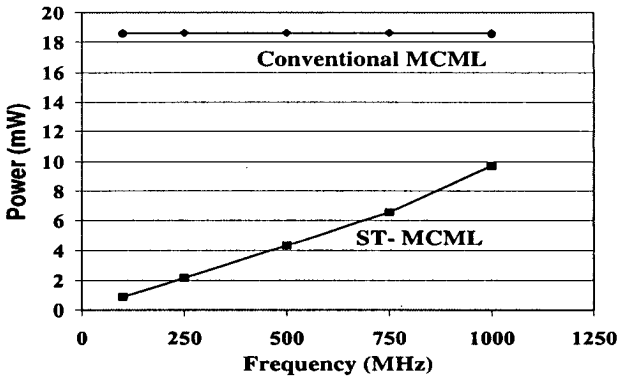


Fig. 8. Power Graph for a 4-bit CLA adder

worst case delay which is produced by setting the input vectors A=0's, B=1's and Cin toggles between "1" and "0".

At an input frequency of 1GHz and supply voltage of 1.8V, ST-MCML offers 50% reduction in PDP over static CMOS and MCML in the case of the 4-bit ripple adder. On the other hand, ST-MCML offers 22% and 56% PDP reduction compared to static CMOS and MCML respectively.

Figure 8 shows as the operating frequency varies, the power dissipation of the conventional MCML design remains constant, while power of the ST-MCML design is proportional to the frequency. Over a wide range of frequencies ST-MCML dissipates significantly less power compared to conventional MCML.

An approximate measurement for the area overhead in ST-MCML designs due to the added pulse generator and sense amplifiers has also been calculated. An increase of 15% of area was calculated for the 4-bit ripple adder case, while only 7% area increase was reported for the 16-bit CLA adder. It could be therefore concluded that the larger the design being implemented in ST-MCML, the less significant an area increase would be. On the other hand, the constant current source in conventional MCML designs reduces the switching noise and supply fluctuations. Therefore, MCML is recommended for mixed signal design to reduce the interference between the digital and analog blocks [2]. But for pure digital circuits like the ones investigated in this work, replacing the constant current source with a pulsing one, would cause minimal noise influence over the digital blocks. However, ST-MCML should not be used in mixed signal designs.

5. CONCLUSION

A Self-Timed MOS Current-Mode Logic (ST-MCML) for digital applications is developed. 4-bit ripple and 16-bit Carry Look Ahead adders are implemented using the ST-MCML technique in a 0.18μm 1.8V 1-GHz CMOS process. ST-MCML is compared to conventional MCML, CMOS and domino logic in terms of power, delay, Power-Delay-Product (PDP) and Energy-Delay-Product (EDP). ST-MCML achieves 50% reduction in PDP over static CMOS and MCML in the case of the 4-bit ripple adder. On the other hand, ST-MCML offers 22% and 56% PDP reduction compared to static CMOS and MCML respectively.

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